# Saurav Dutta

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Personal Page

## Research Interests

Multiscale Multiphysics, Metamaterials, Soft Robotics, Morphing Structures, Continuum Mechanics, Non-linear Dynamics, Bifurcation, Molecular Dynamics Simulation, Wave Propagation and Data-driven Mechanics.

## Education

**National Institute of Technology, Silchar** [Jul '19 - Jul'23]
Bachelor of Technology | Department of Civil Engineering

- Major cumulative grade point average (CPI): **9.03/10**

## Conference Publication

**$8^{th}$ International Conference On Recent Advances in Geotechnical Earthquake Engineering and Soil Dynamics (8ICRAGEE)**(Scheduled) [Dec '24]
**Saurav Dutta**, Vishwajit Anand,"Framework for Ground Motion Characterization"

## Journal Publication

Manish Kumar, **Saurav Dutta**, Srishti and Vishwajit Anand,*'Identification of Critical Ground Motion Features'*(in preparation)

## Scholastic Achievements

- Attained **AA** grade in Bachelors thesis Project I and II in $7^{th}$ and $8^{th}$ semester based on exceptional performance ['23]
- Attained **AA** grade in **17** out of **27** department courses based on exceptional academic performance ['23]
- Scored a **9.93/10** SPI in fourth semester and maintained an average of **9.6+** CPI in the previous six semesters ['23]
- Selected as an Undergraduate Research Student under one of the Undergraduate Research Council Funded Project in $8^{th}$ semester at NIT, Silchar ['23]
- Selected as a Summer Research Intern at **Indian Institute of Technology, Guwahati** in the Department of Mechanical Engineering under Technical Innovation Hub (TIH) ['22]
- Selected as a Summer Research Fellow at **Indian Institute of Technology, Varanasi** in the Department of Civil Engineering ['22]
- Cleared Pre-regional Mathematical Olympiad and was selected for Regional Mathematical Olympiad ['17]

## Research Experience

### Indian Institute of Sciences (IISc), Bangalore, India

*Junior Researcher* | *Guide: Prof. Akshay Joshi* [Aug '24 - Present]

- Involved in the development of experimental setup tpo study wave-propagation in thin films/plates
- Involved in the study of Data driven mechanics to analysis material behaviour.

### Indian Institute of Sciences (IISc), Bangalore, India

*Research Assistant/Project Associate* | *Guide: Prof. Rajesh Chaunsali* [Jun '23 - July '24]

- Developed the experimental setup to study the periodically time varying stiffness in a 1-DOF Active Pendulum System.
- Surveyed literature about the Non-reciprocal Robotic system, Odd Elastodynamics and it's topological invariants.
- Conducted the analytical and numerical analysis of non-reciprocal metamaterial and non-hermitian topological systems

### National Institute of Technology (NIT), Silchar, India

*Bachelors Thesis Project II* | *Guide: Prof. Atanu Sahu* [Jan '23 - May'23]

- Surveyed literature about the state-of-the-art of the analytical and dynamical behaviour of composite plates under different loads using various methods
- Worked on the numerical and analytical implementation of the static and dynamic behavior of laminated composite plate under **thermal effect**
- Implemented the Static and dynamic behavior using **thermal stress** in ABAQUS

### Indian Institute of Technology (IIT), BHU, India

*Summer Research Fellow* | *Guide:Prof. Vishwajit Anand* [May'22 - July'22]

- Surveyed literature about the state-of-the-art of the Ground motion parameters study
- Wrote a generalised code in MATLAB to obtain various parameters of a ground motion
- Characterized those parameters as critical or negligible effects on structures

## Research Projects

### Non-reciprocity in Damped Elastic Periodic Lattices ['23 - Present]

*Guide : Prof. Rajesh Chaunsali| Department of Aerospace Engineering*

- Surveying Literature on Non-hermitian Skin Effect (NHSE) in Mechanical Systems.
- Wrote a code on MATLAB to analyze Non-reciprocity in Spring on a Monomer Chain to do an initial study on *NHSE*
- Obtained the Dispersion relation analytically, numerically, and correlated with the finite chain analysis
- Wrote a generalized code for the 1D infinite and finite chains for combined non-reciprocity in Spring and Damping.

### Experimental Realisation of Periodic Time varying Stiffness in 1-DOF ['23 - '24]

*Guide : Prof. Rajesh Chaunsali| Department of Aerospace Engineering*

- Developed a generalized code in Python for Position and speed control of Multiple Dynamixel servo Motors.
- Designed the Pendulum Model and Circular Disc model in CATIA, 3D Printed the models, and fitted the motors inside.
- Developed the Experimental Setup of a 2 DOF Pendulum System with magnets inside the circular disc, which provides periodic stiffness.
- Developed the Video Processing code in Python using OpenCV to post-process the Time periodic setup data of the two pendulums.
- Experiments include: Damping Calibration, Passive(in phase and out of phase) experiment, and Active(Motor ON) experiment.

### Static and Dynamic Analysis of a Laminated Composite Plate under Thermal Effects using ABAQUS ['23 ]

*Guide : Prof.Atanu Sahu | Department of Civil Engineering*

- Literature Survey on Laminated Composite plates
- Studied the analytical behavior of composites under different environmental conditions.
- Worked on ABAQUS to model a laminated composite plate
- Applied different Boundary Conditions, Loading conditions (Point load), Thermal Stress, Meshing, and Post-processed.
- Analysed the same structure under Static and Dynamic Conditions

### Characterization of Ground Motions ['22]

*Guide: Prof. Vishwajit Anand | Department of Civil Engineering*

- Literature Survey on Ground motions and its parameters.
- Worked on the development of a generalized code concerning an open-source software OpenSeismoMatlab.
- Wrote a generalized code in MATLAB which estimates more than 30 ground motion parameters, also included some new parameters.

## Academic Curriculum Work

### Design and Analysis of G+4 Residential Building ['22]

*Bachelor's Project I | Guide: Prof. Subhrajit Dutta*

- Performed hand calculation of BM and SF of continuous beams, columns, Slabs, and Staircase to estimate the no. of r/f required as per IS Codes.

- Developed the building model in ETABS to do the analysis and obtain the design steel values.
- Performed wind and seismic analysis on the building as per IS Codes.
- Detailing of each section was done manually.

### Stress Analysis using Finite Element Method ['23]

*Course: Finite Element Method | Instructor: Prof. Atanu Sahu*

- Developed a Finite Element code in MATLAB to analyze the truss subjected-to-point loads.
- It also included calculating forces in each element of a truss system.
- Compared the results with the conventional method.

## Relevant Courses Undertaken

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| **Structures** | Solid Mechanics, Structural Analysis, Vibration and Structural Dynamics, Finite Element Method, Plates and Shells, Composite Materials<br>Advanced Structural Analysis, Mechanics of Materials, Wave Propagation in Designed Materials |
| **Mathematics** | Linear Algebra, Partial and Ordinary Differential Equations, Calculus |
| **Miscellaneous** | Control System Theory, Design of Reinforced Concrete Structures, Design of Steel Structures, Remote Sensing and GIS, Fluid Mechanics, Soil Mechanics, Earthquake Engineering, Modeling and Simulation, Basics of Electricity |

## Position Of Responsibility

### Guest Lecturer, Wave Propagation in Designed Materials [Aug '23 - Dec '23]

*Prof. Rajesh Chaunsali | Dept. of Aerospace Engineering, IISc, Bangalore*

- Conducted one class in collaboration with a PhD Student on Recent Advances in the field of Metamaterials.
- Explained how to create active metamaterial by inducing Non-reciprocity in the system.
- Explained the phenomenon *Non-hermitian Skin Effect* from Wave propagation theory.
- Gained initial insights into the teaching assistantship by explaining new concepts in the field of Metamaterials.

## Technical Skills

**Languages** : MATLAB, Python, Mathematica, LaTeX, HTML, CSS, C++, C
**Softwares** : CATIA, ANSYS, COMSOL Multiphysics, AutoCAD, Abaqus, Dynamixel Wizard 2.0
**Experimental** : Motor Control, Embedded Sensors, Arduino, U2D2, Laser Doppler Vibrometer, 3D Printing

## Extracurricular Activities

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|---|---|
| **Social** | • Served as **Head** at the **School Genius** organized by **Tecnoesis NITS** where various age group of high school students were managed ['22]<br>• Served as **Head** at the **Razzmatazz** organized by **Incandescence NITS** and organized various events like quizzes, gaming competition ['23] |

## References


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Dr. Vishwajit Anand
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